

MLPerf Storage v3.0 — System Description

hickory · CLOSED division · submitter: farmgpu

System name	hickory
Submitter	FarmGPU (farmgpu.com)
Division	CLOSED
Benchmark(s)	KVCache, Training (unet3d), Checkpointing — MLPerf Storage v3.0 (mlpstorage 3.0.46 / v3.0-rc1)
Solution type	shared parallel filesystem (POSIX file API), on-premises
Availability	preview
Usable capacity	646.58 TiB — the platform-reported usable capacity after 5+2 erasure coding and reserve, over 1,106.1 TB of raw flash. Submitted as <code>usable_capacity_tib</code> .
Deployment	on-premises, bare metal; WEKA containers orchestrated on Kubernetes
Rack units	16 U storage-under-test (8x 2U backend nodes). Client nodes excluded per the MLPerf RU definition.
Client topology	15 client hosts (8x parsnip + 7x potato) x 2 MPI ranks/host = 30 ranks for KVCache; 512 processes (TP8 x PP32 x DP2) across 15 hosts for Checkpointing; 15 hosts x 75 simulated accelerators for Training
Model(s) exercised	KVCache: llama3.1-8b, llama3.1-70b-instruct · Checkpointing: llama3-405b, llama3-1t · Training: unet3d
Run date(s)	2026-07-24
Document prepared	2026-08-03

1. System overview

The storage under test is a WEKA® Data Platform parallel filesystem cluster of 8 dedicated backend nodes, presenting a shared POSIX namespace to 15 bare-metal benchmark clients over RoCEv2 with the DPDK client transport on a dual-rail 400G fabric. All benchmark I/O traverses the network to the backend; the clients are pure I/O generators and their local NVMe is not part of the solution under test. Non-volatile capacity is provided entirely by Solidigm™ D7-PS1010 7.68 TB TLC PCIe Gen5 NVMe SSDs — 20 fitted per backend node (160 total), of which 144 (18 per node) were claimed by the cluster during the submitted runs. The cluster runs 5+2 erasure coding with no hot-spare failure domain over 1,106.1 TB of raw flash, yielding 646.58 TiB (710.9 TB) of usable capacity as reported by the platform. The filesystem exercised by the benchmark ('default', mounted at /mnt/wk0) was provisioned at 200.0 TiB of that usable capacity.

2. Hardware

Backend node — 8 × (the storage under test)

Chassis	2U — 2 rack units per node, 16 U total
CPU	1 × AMD EPYC™ 9655 — 96 cores, 1 socket
DRAM	1536 GiB installed — 24 × 64 GB DDR5-4000 (Samsung M321R8GA0EB2-CCPKC); 1511 GiB available to the OS
Storage under test	20 × Solidigm™ D7-PS1010 7.68 TB TLC PCIe Gen5 NVMe (P/N SB5PH27X076T) fitted per node, 160 total; 144 (18 per node) ACTIVE in the cluster during the submitted runs
Data NIC	2 × NVIDIA® ConnectX®-8 dual-400GbE, RoCEv2 — one port group per storage fabric (A and B)
Management NIC	1 × 1 GbE, out-of-band management network
Power supplies	2 × 2000 W Titanium (240 V); min PSUs active = 1 (1 redundant)
Storage software	WEKA® Data Platform 5.1.2.420-cx8

parsnip client — 8 × (not part of the storage under test)

Chassis	Supermicro SYS-212H-TN, 2U
CPU	1 × Intel® Xeon® 6781P — 80 cores / 160 threads
DRAM	1024 GiB installed; 1007 GiB available to the OS
Data NIC	4 × 400 GbE NVIDIA® ConnectX®-8, RoCEv2, dual-rail (net:ha) mount
Management NIC	1 × 10 GbE
Local NVMe	16 × Solidigm™ D7-PS1010 15.36 TB TLC (P/N SB5PH27X153T) — client-local, not under the control of the storage system and excluded from reported capacity
Operating system	Ubuntu 24.04

potato client — 7 × (not part of the storage under test)

Chassis	Supermicro SYS-212H-TN, 2U
CPU	1 × Intel® Xeon® 6781P — 80 cores / 160 threads
DRAM	512 GiB installed — 8 × 64 GB DDR5-5600 (Micron MTC36F2046S1PC56BG1); 503 GiB available to the OS
Data NIC	2 × 400 GbE NVIDIA® ConnectX®-7, RoCEv2, dual-rail (net:ha) mount
Management NIC	1 × 1 GbE
Local NVMe	24 × Solidigm™ D5-P5336 122.88 TB QLC (P/N SBFPF2BV0P12) — client-local, excluded from reported capacity
Operating system	Ubuntu 24.04

Client power supplies are omitted: §11.4.2 asks for the hardware required to operate the storage system, which excludes the benchmark clients. See §5 for the power table.

3. Software

Storage software	WEKA® Data Platform 5.1.2.420-cx8 (container image weka-in-container-farmgpu:5.1.2.420-cx8), operator-managed on Kubernetes; 8 compute + 8 drive containers across 8 hosts
Filesystem / protocol	WEKA shared parallel filesystem 'default', POSIX file API, mounted at /mnt/wk0; distributed metadata, stateless clients, erasure-coded data protection
Client transport	DPDK with RoCEv2, dual-rail mount (net:ha), MTU 9000
Client operating system	Ubuntu 24.04
Benchmark	MLCommons mlpstorage 3.0.46 (v3.0-rc1)
Launcher	Open MPI (mpirun)

4. Settings and non-default configuration

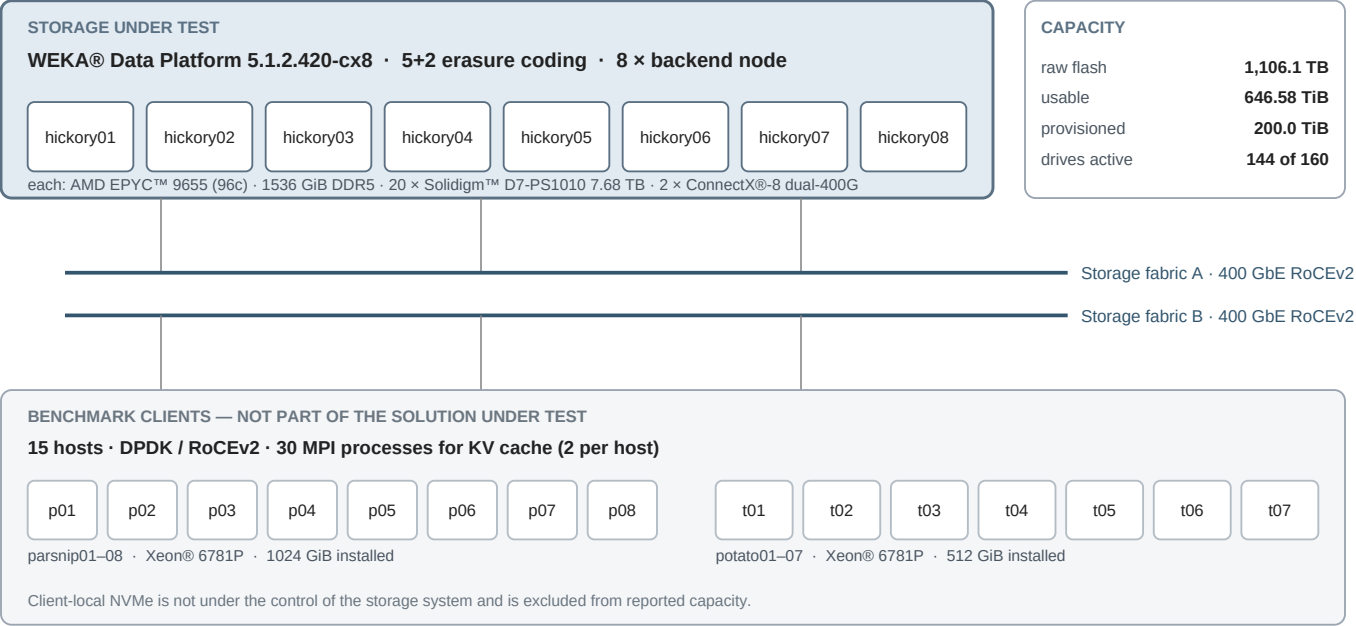
- Client mount: WEKA POSIX client over DPDK with RoCEv2, dual-rail (net:ha) across both storage fabrics simultaneously, MTU 9000. KV cache directory /mnt/wk0/kvcache; results directory /mnt/mlperf_results on a separate filesystem, verified by the harness fs-separation probe (skip_fs_separation_gate = false).
- Client-side invocation parameters for every run on this system are recorded verbatim in the executed_command field of each run's metadata in the submitted results tree.
- KV cache topology: 30 MPI processes across the 15 client hosts (2 per host). Harness settings: 3 trials per option, seed 42, 90 s inter-option delay, autoscaling enabled in qos mode, RAG enabled with 10 documents per query, prefix caching enabled. --skip-validation was NOT used; all harness pre-run probes were active (skip_validation = false, skip_ssh_check = false).
- Per-option concurrency is fixed by the CLOSED workload definition, not chosen by us: Option 1 at --num-users 200, Option 2 at --num-users 100, Option 3 at --num-users 70. The reported ordering, in which Option 1 exceeds Option 2 on read bandwidth, follows from that 2x concurrency difference together with Option 2's DRAM tier (--cpu-mem-gb 4) routing the majority of read volume away from storage. Each option leads on the metric it was designed for. This was raised as GitHub issue #131 in the submission repository and closed on 2026-08-04.
- Cluster state during the submitted runs: 144 of the 160 fitted Solidigm SSDs were ACTIVE in the cluster (18 per node, uniformly); the remaining 16 were fitted but unclaimed. The benchmark filesystem was provisioned at 200.0 TiB of the 646.58 TiB usable capacity, with thin provisioning disabled.
- Training and Checkpointing on this system used a separate client configuration from the KV cache runs. Training ran unet3d across 15 hosts with 75 simulated accelerators over a dataset of 553,166 files / 81.09 TB; Checkpointing ran 512 processes (TP8 x PP32 x DP2) across 15 hosts with fsync enabled.

5. Power requirements

System component	Power supply unit	Nameplate rated power	Design power
Backend storage node x8	PSU 1	2000 W	2000 W
	PSU 2 (redundant)	2000 W	—
Totals		8 x 4000 W = 32,000 W	8 x 2000 W = 16,000 W

Design power is stated at one active PSU per node (redundant configuration, min PSUs active = 1); nameplate totals count both PSUs. Only the 8 backend nodes appear here: §11.4.2 asks for the hardware required to operate the storage system, which excludes the 15 benchmark client nodes and the client fabric switching. Values are provisioned power derived from PSU ratings, not measured consumption. PSU RATING UNCONFIRMED — the 2000 W Titanium figure is representative and pending confirmation against the chassis datasheet or BMC inventory.

6. System topology



All benchmark I/O crosses the 400G fabric to the WEKA backend. Both storage fabrics are used simultaneously by clients via the dual-rail (net:ha) mount. Clients hold no part of the storage solution.

7. Rack units

Per the MLPerf Storage RU definition, total_rack_units counts the storage-under-test plus required backend gear and excludes client nodes. The 8 backend nodes occupy 2 U each, so total_rack_units = 16, which is what the systems YAML records. The 15 client nodes and the client-side fabric switching are excluded.

Generated for the MLPerf Storage v3.0 farmgpu submission from systems/hickory.yaml. Hardware values correspond to the system YAML; capacity is as measured. Prepared 2026-08-03.